

Causal Inference With Network Data

Empirical Methods — Week 19

Teaching deck draft

Week 19 question

- How can researchers estimate effects when units influence one another through observed or latent links?
- The problem is causal inference, not general network analysis.
- Network structure changes identification, estimation, inference, and interpretation.
- The graph definition is substantive: it states who can affect whom and through which channel.

Lecture 18 built the graph. Lecture 19 asks what can be learned causally once the graph exists.

Why Lectures 1–5 are not always enough

Core design habit	Network complication	Response
Individual treatment contrast SUTVA notation	untreated units may be exposed outcomes depend on others' treatment	define exposure states potential outcomes under inter- ference
Peer regression	outcomes jointly determine each other	topology, timing, randomiza- tion
One row per unit Cluster standard errors	rows are dyads or links dependence follows shared nodes	dyadic/network inference pair-aware variance logic

The core toolkit still matters; the network tells us when its assumptions are too small.

Potential outcomes under interference

Standard binary-treatment notation:

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Network notation:

$$Y_i = Y_i(Z_i, Z_{-i}; G)$$

Exposure-mapping restriction:

$$D_i = g_i(Z_{-i}, G), \quad Y_i = Y_i(Z_i, D_i)$$

- G : graph of economically relevant links.
- Z_i : own treatment status.
- D_i : exposure to others' treatments through the graph.
- The mapping $g_i(\cdot)$ is an identifying assumption, not a formatting choice.

The reflection problem

$$Y_i = \alpha + \beta \sum_j w_{ij} Y_j + \gamma \sum_j w_{ij} X_j + \delta X_i + \varepsilon_i$$

- β : endogenous peer effect.
- γ : contextual peer-characteristic effect.
- Correlated effects: sorting, common shocks, shared managers, shared neighborhoods.
- Peer outcomes and own outcomes can be jointly determined.
- Leave-one-out peer means reduce mechanical self-inclusion; they do not solve sorting.

Network topology helps only when the exclusion logic is economically credible.

Interference and exposure mappings

$$\tau^{direct}(d) = \mathbb{E}[Y_i(1, d) - Y_i(0, d)]$$

$$\tau^{spill}(z; d, d') = \mathbb{E}[Y_i(z, d) - Y_i(z, d')]$$

- Direct effects compare own treatment at a fixed exposure state.
- Spillover effects compare exposure states at a fixed own treatment.
- Exposure can be treated-neighbor share, treated manager, treated high-centrality contact, or saturation cell.
- Failure mode: the exposure mapping misses the true path or compresses heterogeneous paths too aggressively.

Randomized saturation

$$Z_i \sim \text{Bernoulli}(p_c), \quad p_c \in \{p_L, p_H\}$$

- Randomize treatment intensity across teams, classrooms, villages, worksites, or graph partitions.
- Untreated units in high-saturation clusters reveal spillovers relative to untreated units in low-saturation clusters.
- Treated units across saturation cells reveal whether direct effects depend on peer exposure.
- Partial interference assumes spillovers stay inside the cluster.
- Strong design, but fragile if spillovers cross the boundary.

Network-targeted econometric methods

Tool	Identifies or protects	Main caveat
Topology-based peer design	peer effects under excluded-peer variation	network and exclusion restrictions
Exposure mapping	effects of direct and indirect exposure states	mapping must match mechanism
Saturation experiment	direct, spillover, intensity effects	boundaries must be plausible
Randomization inference	design-based p-values	must follow assignment rule
Leave-one-out peers	cleaner peer exposure measure	not a full design
Dyadic inference	valid pair-level uncertainty	does not solve link endogeneity

Dyadic inference

Dyadic model:

$$Y_{ij} = X'_{ij}\beta + a_i + b_j + u_{ij}$$

- Observations are pairs: worker–firm, employee–applicant, worker–worker, sender–receiver.
- Dyads sharing i can be correlated.
- Dyads sharing j can be correlated.
- IID or ordinary robust standard errors usually understate uncertainty.
- Pair-aware inference clusters or otherwise accounts for shared-node dependence.

The estimating unit and variance formula must match the relational data structure.

Network definition changes interpretation

Link	Possible mechanism	Interpretation risk
Coworker	learning, norms, monitoring, team production	same firm may just be sorting
Manager	supervision, information, practices	manager assignment may be endogenous
Referral	information, screening, trust, favoritism	tie may proxy applicant quality
Neighborhood	job information, local norms, commute knowledge	proximity may proxy local shocks
Platform interaction	congestion, reputation, matching	platform rules shape observed links

The graph is part of the estimand, not just an input file.

- Manski: reflection problem and endogenous social effects.
- Bramouille, Djebbari, Fortin: network topology and peer-effect identification.
- Aronow and Samii: causal effects under general interference.
- Athey, Eckles, Imbens: exact inference with network interference.
- Cornelissen, Dustmann, Schonberg: workplace peer effects.
- Pallais and Sands: referrals and field-experimental labor-market design.
- Graham and Canen: dyadic/network data and dependence-aware inference.

Reproduce

- synthetic teams
- coworker links
- treatment saturation
- leave-one-out peers
- spillover estimates

Diagnose

- exposure mappings
- reflection risk
- randomization inference
- partial interference
- common shocks

Transfer

- referral dyads
- pair-level model
- naive vs dyadic SE
- access inequality
- design memo

What each method identifies

Method	Target	Not automatically identified
Peer-effect regression	influence within a defined group	formation or common shocks
Exposure mapping	effect of exposure state d	all paths in the network
Saturation design	spillovers within design cells	market-wide equilibrium effects
Network experiment	effect under assignment protocol	effects under a different graph
Dyadic model	pair-level association or effect	why the link exists

Takeaways

- Network causal inference starts by defining who can affect whom.
- Reflection separates peer correlations from peer effects.
- Exposure mappings turn impossible potential-outcome objects into estimands.
- Saturation and graph experiments randomize exposure, not only own treatment.
- Partial interference is an assumption about boundaries.
- Dyadic data need dependence-aware inference.
- Lecture 20 asks what happens when the network itself is endogenous.